

Self-powered textiles: enhancements to the piezoelectric properties of polyvinylidene fluoride and quartz on fabric using copper nanoparticles

Introduction

- Piezoelectric material is able to create electrical pulses when put under mechanical stress (Latif et al., 2024)
- Polyvinylidene Fluoride (PVDF) and quartz (SiO₂) have been found to be piezoelectric due to their crystalline properties and have been researched in flexible environments (Wang et al., 2025)
- Conductive additives to piezoelectric materials have been found to increase the electrical output of such materials (Almussalam et al., 2017)
- Can be potentially applied to smartwatches, powered shoes and even larger pieces of fabrics
- Little research has been previously on combined piezoelectric materials

OBJECTIVE: This research aims to enhance the piezoelectric properties of PVDF and SiO₂ deposited onto textiles using copper nanoparticles (CuNP)

HYPOTHESIS: as the CuNP concentration increases, the electric output will first increase and then decrease, as a similar trend was observed with Lead Zirconate Titanate, the most common (but toxic) piezoelectric material (Almussalam et al., 2017)

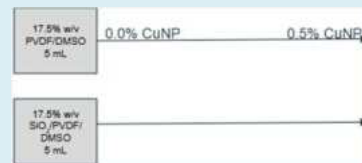
VARIABLES:

IV: CuNP concentration in w/v

DV: Output voltage in Volts

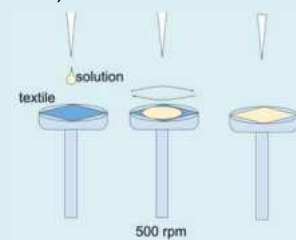
Control: Textiles without added CuNP

Methodology

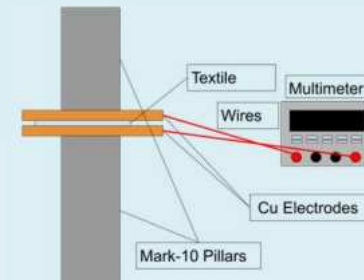


1. Solutions: 5 mL of each solution (0, 0.1, 0.2, 0.3, 0.4, 0.5)

2. Spin Coating: 1 mL of each solution on each textiles (30 textiles for each batch, 60 textiles total)



4. Graphs: Calculating means and standard deviations of each textile -> calculating total mean and standard error -> graphing



3. Testing: 0-12 N compression on textiles, 10 trials for each textiles

Results

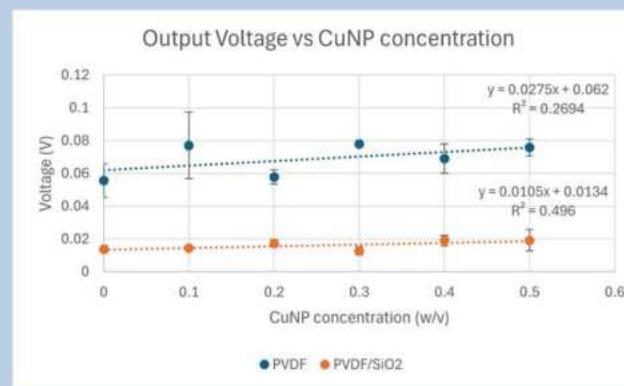


Figure 1. Graph of output voltage vs CuNP concentration



Figure 4. Regular Textile Morphology



Kruskal Wallis Test (5% Significance)

PVDF/DMSO:

Df = 5

P = 0.66

H-statistic = 3.23

PVDF/SiO₂/DMSO:

Df = 5

P = 0.55

H-statistic = 3.99

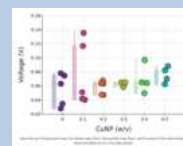


Figure 3. Distribution of PVDF

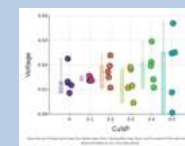


Figure 2. Distribution of PVDF/SiO₂

Figure 5. Textile Morphology with deposited Solution

Conclusion

Based on the Kruskal Wallis test for both groups and Figure 2 and 3 that show the distribution of the data, it seems that there is little to no significant difference between the output voltage of different concentrations of CuNP. This can be concluded because the P values for both tests can be more than 0.05, which means that we fail to reject the null hypothesis that state the groups are not equal. Based on the graph (Figure 1), the CuNP has a positive correlation with output voltage for both groups, though the correlation is much more significant in the PVDF/DMSO solution than it is in the PVDF/SiO₂/DMSO solution, indicated by the higher incline. Also, based on Figure 1, it seems that the output voltage of PVDF/DMSO is higher than that of PVDF/DMSO/SiO₂. Based on the results of the experiment, the initial hypothesis was refuted, as not only was there an exclusively upwards trend for both, but the Kruskal Wallis test shows that the data supports that the CuNP concentration did not cause a significant difference in the output voltage. A large source of error is the method of spin coating. Despite using constant rpm and solution amount, it is not confirmed if the solution is equally distributed on each of the textiles, or that the distribution of each of the textiles is the same. If they are different, especially if not all the solution deposited onto the textile, it results in lower piezoelectric material on the textile, therefore leading to lower output voltages. Inconsistencies in applying force may have also caused error.

Acknowledgements

Almussalam, A., Luo, Z., Komolafe, A., Yang, K., Robinson, A., Torah, R., & Beeby, S. (2017). Flexible piezoelectric nano-composite films for kinetic energy harvesting from textiles. *Nano Energy*, 33, 140-156. <https://doi.org/10.1016/j.nanoen.2017.10.037>

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